Laboratory work 4

Adding some values for tasks:

Query | Query History

```sql
INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price)
VALUES (201, (SELECT min(flight_id) FROM flights), 1, 'Website',     '2024-04-10', now(), 'Confirmed', 45000.00),
       (250, (SELECT min(flight_id) FROM flights), 2, 'Mobile App',  '2024-04-12', now(), 'Pending',   52000.00),
       (300, (SELECT min(flight_id) FROM flights), 3, 'Agency',      '2024-04-15', now(), 'Confirmed', 61000.00);

INSERT INTO baggage_check (check_result, created_at, updated_at, booking_id, passenger_id)
VALUES ('Not checked', '2024-04-11 09:00', '2024-04-11 09:00', 201, 1),
       ('Pending',     '2024-04-13 10:30', '2024-04-13 10:30', 250, 2),
       ('Checked',     '2024-04-16 08:15', '2024-04-16 08:15', 300, 3);

INSERT INTO baggage_check (check_result, created_at, updated_at, booking_id, passenger_id)
VALUES ('Not checked', '2024-05-20 14:00', '2024-05-05 09:00', 201, 1),
       ('Checked',     '2024-06-25 16:30', '2024-06-02 11:45', 250, 2);
```

Data Output | Messages | Notifications

```
INSERT 0 2

Query returned successfully in 237 msec.
```

Query returned successfully in 237 msec.

Total rows: | Query complete 00:00:00.237 | CRLF | Ln 13, Col 72

Task 1

Query | Query History

```sql
SELECT airline_id, airline_name AS original_name, UPPER(airline_name) AS name_uppercase FROM airline ORDER BY airline_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 24 | Page No: 1 of 1

	airline_id [PK] integer	original_name character varying (50)	name_uppercase text
1	1	Air Astana	AIR ASTANA
2	2	SCAT	SCAT
3	3	FlyArystan	FLYARYSTAN
4	4	Qazaq Air	QAZAQ AIR
5	5	Global Airways	GLOBAL AIRWAYS
6	6	Lufthansa	LUFTHANSA
7	7	Emirates	EMIRATES
8	8	Air France	AIR FRANCE
9	9	LOT	LOT
10	10	Aeroflot	AEROFLOT
11	11	Wizz Air	WIZZ AIR
12	12	Ryanair	RYANAIR
13	13	Qatar Airways	QATAR AIRWAYS
14	14	KLM	KLM
15	15	Pegasus	PEGASUS
16	16	Uzbekistan Airways	UZBEKISTAN AIRWA...
17	17	China Southern	CHINA SOUTHERN
18	18	Korean Air	KOREAN AIR
19	19	Etihad	ETIHAD
20	20	Finnair	FINNAIR
21	21	KazAir	KAZAIR

Total rows: 24 | Query complete 00:00:00.155 | CRLF | Ln 1, Col 102

Task 2

Query | Query History

```
1  select airline_id, airline_name,
2  replace(airline_name, 'Air', 'Aero') as replaced_name from airline where airline_name like '%Air%'
3  order by airline_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	airline_id [PK] integer	airline_name character varying (50)	replaced_name text
1	1	Air Astana	Aero Astana
2	4	Qazaq Air	Qazaq Aero
3	5	Global Airways	Global Aeroways
4	8	Air France	Aero France
5	11	Wizz Air	Wizz Aero
6	13	Qatar Airways	Qatar Aeroways
7	16	Uzbekistan Airways	Uzbekistan Aerowa...
8	18	Korean Air	Korean Aero
9	21	KazAir	KazAero
10	22	AirEasy	AeroEasy

Task 3

Query | Query History

```
1  select flight_id from flights where airline_id = 1
2  intersect select flight_id from flights where airline_id = 2;
```

Data Output | Messages | Notifications

flight_id integer

No intersection, because each flight connects to only one airline

Task 4

Query | Query History

```
1  select * from airport where airport_name ilike '%Reginal%' or airport_name ilike '%Air%'
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	3	Pavlodar Airport	Kazakhstan	Pavlodar Region	Pavlodar	2023-05-27 00:00:00	2026-09-29 03:16:13.89957
2	11	Chopin Airport	Poland	Masovia	Warsaw	2023-08-28 00:00:00	2026-09-29 03:16:13.89957
3	14	Istanbul Airport	Turkey	Istanbul Province	Istanbul	2023-03-17 00:00:00	2026-09-29 03:16:13.89957
4	19	Shymkent Airport	Kazakhstan	Turkestan Region	Shymkent	2023-02-13 00:00:00	2026-09-29 03:16:13.89957
5	20	Sary-Arka Airport	Kazakhstan	Karaganda Region	Karaganda	2023-09-23 00:00:00	2026-09-29 03:22:10.536374

Task 5

Query | Query History

```
1  select * from passengers;
2  select first_name, last_name, to_char(date_of_birth, 'FMMonth DD, YYYY') as birth_date_formatted from passengers order by passenger_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 50 | Page No: 1 of 1

	first_name character varying (50)	last_name character varying (50)	birth_date_formatted text
1	Anna	Petrov	July 02, 1991
2	Aliya	Kim	February 09, 1964
3	Timur	Smagulov	June 01, 2002
4	Madina	Abenov	December 08, 1968
5	Yerlan	Li	March 27, 1977
6	Madina	Petrov	March 05, 1965
7	Timur	Abenov	March 14, 1960
8	Olga	Li	July 13, 1985
9	Dana	Zhaksy	March 04, 1988
10	Nurlan	Seitova	October 01, 1983
11	Aliya	Ivanova	February 04, 1990
12	Ivan	Seitova	December 09, 1981
13	Ivan	Li	September 10, 1961
14	Madina	Li	September 11, 1994
15	Madina	Li	June 24, 1991
16	Aliya	Ivanova	December 12, 1988
17	Madina	Ivanova	August 20, 1973
18	Yerlan	Zhaksy	March 14, 1990
19	Ivan	Petrov	January 05, 1968
20	Nurlan	Li	January 01, 1968
21	Aidar	Abenov	May 17, 1973

Task 6

Query | Query History

```
1  select flight_id, sch_departure_time, act_departure_time,
2  extract(epoch from (act_departure_time - sch_departure_time)) / 60 as delay_minutes from flights
3  where act_departure_time > sch_departure_time order by delay_minutes desc;
4
```

Data Output | Messages | Notifications

Showing rows: 1 to 19

	flight_id [PK] integer	sch_departure_time timestamp without time zone	act_departure_time timestamp without time zone	delay_minutes numeric
1	25	2023-02-10 16:00:00	2023-02-10 16:47:00	47.0000000000000000
2	17	2025-08-14 04:00:00	2025-08-14 04:45:00	45.0000000000000000
3	22	2025-01-03 12:00:00	2025-01-03 12:43:00	43.0000000000000000
4	26	2023-01-18 09:00:00	2023-01-18 09:39:00	39.0000000000000000
5	23	2023-03-17 03:00:00	2023-03-17 03:39:00	39.0000000000000000
6	15	2025-12-24 08:00:00	2025-12-24 08:35:00	35.0000000000000000
7	28	2023-05-08 13:00:00	2023-05-08 13:33:00	33.0000000000000000
8	35	2025-03-14 15:00:00	2025-03-14 15:30:00	30.0000000000000000
9	30	2025-04-08 14:00:00	2025-04-08 14:24:00	24.0000000000000000
10	19	2025-07-24 16:00:00	2025-07-24 16:23:00	23.0000000000000000
11	33	2025-03-12 07:30:00	2025-03-12 07:50:00	20.0000000000000000
12	24	2023-10-02 09:00:00	2023-10-02 09:18:00	18.0000000000000000
13	14	2025-08-02 17:00:00	2025-08-02 17:18:00	18.0000000000000000
14	31	2025-03-10 08:00:00	2025-03-10 08:15:00	15.0000000000000000
15	13	2025-09-18 06:00:00	2025-09-18 06:15:00	15.0000000000000000
16	16	2023-03-21 05:00:00	2023-03-21 05:14:00	14.0000000000000000
17	11	2023-06-14 08:00:00	2023-06-14 08:13:00	13.0000000000000000
18	29	2025-11-13 06:00:00	2025-11-13 06:10:00	10.0000000000000000
19	32	2025-03-11 09:00:00	2025-03-11 09:05:00	5.0000000000000000

Task 7

Query | Query History

```
1  select f.flight_id, al.airline_name, dep.airport_name as departing_airport, arr.airport_name as arriving_airport,
2  f.sch_arrival_time, f.act_arrival_time,
3  extract(epoch from (f.act_arrival_time - f.sch_arrival_time)) / 60 as delay_minutes from flights f
4  join airline al on al.airline_id = f.airline_id join airport dep on dep.airport_id = f.departing_airport_id
5  join airport arr on arr.airport_id = f.arriving_airport_id
6  where act_arrival_time > sch_arrival_time order by delay_minutes desc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 18 | Page No: 1

	flight_id integer	airline_name character varying (50)	departing_airport character varying (50)	arriving_airport character varying (50)	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone	delay_minutes numeric
1	35	FlyArystan	Frankfurt am Main	Almaty International	2025-03-14 19:00:00	2025-03-14 19:55:00	55.0000000000000000
2	24	Qazaq Air	Frankfurt am Main	Tashkent International	2023-10-02 13:00:00	2023-10-02 13:50:00	50.0000000000000000
3	31	Air Astana	Almaty International	Pavlodar Airport	2025-03-10 12:00:00	2025-03-10 12:40:00	40.0000000000000000
4	33	SCAT	Almaty International	Frankfurt am Main	2025-03-12 11:30:00	2025-03-12 12:10:00	40.0000000000000000
5	22	Air France	Dubai International	Charles de Gaulle	2025-01-03 16:00:00	2025-01-03 16:39:00	39.0000000000000000
6	19	Uzbekistan Airways	Frankfurt am Main	Sheremetyevo	2025-07-24 20:00:00	2025-07-24 20:34:00	34.0000000000000000
7	29	Emirates	Helsinki-Vantaa	Pavlodar Airport	2025-11-13 10:00:00	2025-11-13 10:28:00	28.0000000000000000
8	26	Air France	Chopin Airport	Charles de Gaulle	2023-01-18 13:00:00	2023-01-18 13:27:00	27.0000000000000000
9	11	Aeroflot	Heathrow	Astana International	2023-06-14 12:00:00	2023-06-14 12:27:00	27.0000000000000000
10	30	Finnair	Schiphol	Heathrow	2025-04-08 18:00:00	2025-04-08 18:26:00	26.0000000000000000
11	15	Qatar Airways	Schiphol	Helsinki-Vantaa	2025-12-24 12:00:00	2025-12-24 12:25:00	25.0000000000000000
12	28	KLM	Incheon International	Astana International	2023-05-08 17:00:00	2023-05-08 17:24:00	24.0000000000000000
13	16	Pegasus	Helsinki-Vantaa	Incheon International	2023-03-21 09:00:00	2023-03-21 09:23:00	23.0000000000000000
14	25	Finnair	Hamad International	Frankfurt am Main	2023-02-10 20:00:00	2023-02-10 20:22:00	22.0000000000000000
15	17	Korean Air	Charles de Gaulle	Pavlodar Airport	2025-08-14 08:00:00	2025-08-14 08:22:00	22.0000000000000000
16	14	Aeroflot	Incheon International	Istanbul Airport	2025-08-02 21:00:00	2025-08-02 21:20:00	20.0000000000000000
17	23	Global Airways	Tashkent International	Astana International	2023-03-17 07:00:00	2023-03-17 07:15:00	15.0000000000000000
18	13	Global Airways	Istanbul Airport	Incheon International	2025-09-18 10:00:00	2025-09-18 10:13:00	13.0000000000000000

Task 8

Query | Query History

```
1  select airline_id, airline_name, airline_country, created_at from airline
2  where upper(airline_country) in ('FRANCE', 'PORTUGAL', 'POLAND') and created_at between date '2023-11-01' and date '2024-03-31'
3  order by created_at;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	airline_id [PK] integer	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone
1	8	Air France	FRANCE	2024-02-14 00:00:00

Task 9

Query | Query History

```
1  select bag.baggage_id, bag.updated_at, p.first_name, p.last_name, bag.weight_in_kg from baggage bag
2  join booking book on book.booking_id = bag.booking_id join passengers p on p.passenger_id = book.passenger_id
3  where bag.weight_in_kg > 25
4  order by bag.weight_in_kg desc
5  limit 3;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No:

	baggage_id integer	updated_at timestamp without time zone	first_name character varying (50)	last_name character varying (50)	weight_in_kg numeric (4,2)
1	5	2026-09-29 03:34:12.038554	Aliya	Ivanova	32.50
2	7	2026-09-29 03:34:12.038554	Olga	Seitova	29.27
3	2	2026-09-29 03:34:12.038554	Timur	Smagulov	25.73

Task 10

Query | Query History

```
1  select first_name, last_name, date_of_birth from passengers where date_of_birth = (select max(date_of_birth) from passengers);
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	first_name character varying (50)	last_name character varying (50)	date_of_birth date
1	Madina	Petrov	2005-07-17

Task 11

Query | Query History

```
1  select min(ticket_price) as cheapest_price, booking_platform from booking group by booking_platform order by cheapest_price;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1

	cheapest_price numeric	booking_platform character varying (50)
1	17448.08	Agency
2	17510.25	Website
3	27316.19	Mobile App
4	33215.29	Airport Desk

Task 12

Query | Query History

```
1  select airline_id, airline_name, airline_code, airline_country from airline where airline_code ~ '[0-9]' order by airline_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 18 | Page No: 1 of

	airline_id [PK] integer	airline_name character varying (50)	airline_code character varying (30)	airline_country character varying (50)
1	1	Air Astana	A001	KAZAKHSTAN
2	2	SCAT	A002	KAZAKHSTAN
3	3	FlyArystan	A003	KAZAKHSTAN
4	4	Qazaq Air	A004	KAZAKHSTAN
5	5	Global Airways	A005	United Kingdom
6	6	Lufthansa	A006	GERMANY
7	8	Air France	A008	FRANCE
8	9	LOT	A009	POLAND
9	10	Aeroflot	A010	RUSSIA
10	11	Wizz Air	A011	HUNGARY
11	12	Ryanair	A012	IRELAND
12	13	Qatar Airways	A013	QATAR
13	15	Pegasus	A015	TURKEY
14	16	Uzbekistan Airways	A016	UZBEKISTAN
15	17	China Southern	A017	CHINA
16	18	Korean Air	A018	SOUTH KOREA
17	19	Etihad	A019	UAE
18	20	Finnair	A020	FINLAND

Task 13

Query | Query History

```
1  select airline_id, airline_name, airline_code, airline_country, created_at from airline order by created_at desc limit 5;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1

	airline_id [PK] integer	airline_name character varying (50)	airline_code character varying (30)	airline_country character varying (50)	created_at timestamp without time zone
1	24	FlyFly	FFL	POLAND	2026-09-29 03:51:25.084125
2	22	AirEasy	AEY	FRANCE	2026-09-29 03:51:25.084125
3	23	FlyHigh	FHG	BRAZIL	2026-09-29 03:51:25.084125
4	21	KazAir	KZA	TURKEY	2026-09-29 03:49:18.172309
5	15	Pegasus	A015	TURKEY	2024-11-19 00:00:00

Task 14

Query | Query History

```
1  select baggage_check_id, booking_id, check_result, created_at from baggage_check
2  where booking_id between 200 and 300 and check_result <> 'Checked' order by booking_id;
```

Data Output | Messages | Notifications

	baggage_check_id [PK] integer	booking_id integer	check_result character varying (50)	created_at timestamp without time zone
1	12	201	Not checked	2024-04-11 09:00:00
2	15	201	Not checked	2024-05-20 14:00:00
3	13	250	Pending	2024-04-13 10:30:00

Task 15

Query | Query History

```
1  select baggage_check_id, booking_id, check_result, created_at, updated_at from baggage_check
2  where date_trunc('month', updated_at) = date_trunc('month', created_at) and updated_at < created_at
3  order by created_at;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1

	baggage_check_id [PK] integer	booking_id integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	15	201	Not checked	2024-05-20 14:00:00	2024-05-05 09:00:00
2	16	250	Checked	2024-06-25 16:30:00	2024-06-02 11:45:00